

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<think>The anatomical snuffbox region includes the tendons of the thumb muscles , the superficial palmar arch , and the superficial branch of the radial artery . If a penetrating knife wound is present in this region , it is likely to affect one of the two terminal branches of the radial artery , which are the superficial palmar arch (composed of the terminal branch of the radialis intermediate , the superficial palmar digital branches , and the palmar digital terminal branch of the posterior interosseous artery).</think><answer> Superficial palmar digital branches </answer>

